AERIAL INSIGHTS

PROJECT NAME:	Aerial Insight- Dam Survey
PROJECT GROUP:	Tejas Kale

WHY THIS PROJECT ?

- To understand the use of UAV technology for infrastructure inspection.
- To detect cracks and structural damage in dams safely and efficiently.
- To reduce manual inspection risks in hard-to-reach areas.
- To gain practical experience in drone surveying and data collection.
- To explore real-world engineering applications of drones.

INTRODUCTION

- Used UAV (drone) technology for inspection and monitoring of dam structures.
- Aimed to detect cracks and structural defects safely and efficiently.
- Aerial survey allows inspection of hard-to-reach areas without manual risk.
- Helps in early detection of damage and preventive maintenance.
- Provides accurate visual data for structural analysis.

KEY FEATURES

- UAV-Based Inspection – Drone used for safe and efficient dam monitoring.
- High-Resolution Imaging – Captures clear images for crack detection.
- Access to Difficult Areas – Inspects locations that are hard to reach manually.
- Fast Survey Process – Covers large dam areas in less time.
- Accurate Visual Data – Provides reliable information for structural analysis.

ARCHITECTURE

- UAV Platform – Drone used to fly over the dam and collect data.
- Camera System – Captures images and videos of dam surfaces.
- Control System – Transmitter and flight controller used to operate the UAV.
- Data Storage – Images and videos stored for inspection and study.
- Analysis Method – Collected data used to identify cracks and defects.

RESULTS

- Successfully conducted aerial survey of the dam using UAV technology.
- Captured clear images and videos for crack detection.
- Identified visible cracks and surface defects in the structure.
- Achieved safe and efficient inspection without manual risk.
- Gained practical experience in drone surveying and inspection techniques.

FUTURE SCOPE

1. ☐ **Integrate thermal or advanced cameras for deeper structural analysis.**
2. ☐ **Apply UAV inspection to bridges, buildings, and other strutures.**
3. ☐ **Develop automated flight paths for regular dam monitoring.**